

SIMATS ENGINEERING

ASSESSMENT TOOL 3: Debugging

COURSE NAME: Query Processing for Data Science **COURSE CODE:** DSA0503

COURSE OUTCOME COVERED

CO2: Use Python basics and SQL libraries to connect to databases SQLite, MySQL, PostgreSQL and perform CRUD operations like Create, Read, Update, Delete. (BTL6)

Assessment Tool Weightage: 25%

Question Count: 10 **Total Marks: 20**

Code of Conduct

I _____ S. V. K. Sharon Surya(192324246) __ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: S. V. K. Sharon Surya _____

Debugging

Criteria	Problem Identification (4 Marks)	Debugging (4 Marks)	Optimization (4 Marks)	Analysis (4 Marks)	Code Quality (4 Marks)
Marks					

Total Marks Awarded:

Question 1: SQLite Database Connection

CODE:

```
import sqlite3
```

```
conn = sqlite3.connect("student.db")
```

```
cursor = conn.cursor()

cursor.execute("""
CREATE TABLE IF NOT EXISTS Student(
    Student_ID INTEGER PRIMARY KEY,
    Name TEXT,
    Department TEXT
)
""")

conn.commit()
conn.close()

print("Table Created Successfully")
```

Output

Table Created Successfully

Question 2: SQLite Record Insertion

```
import sqlite3

conn = sqlite3.connect("college.db")
cursor = conn.cursor()

cursor.execute("DROP TABLE IF EXISTS Student")

cursor.execute("""
CREATE TABLE Student(
    Student_ID INTEGER PRIMARY KEY,
    Name TEXT,
```

```
        Department TEXT
    )
    """

cursor.execute("""
INSERT INTO Student(Student_ID, Name, Department)
VALUES(101,'Alice','CSE')
""")

conn.commit()

print("Record Inserted Successfully")

conn.close()
```

Output

Record Inserted Successfully

Question 3: SQLite Data Retrieval

```
import sqlite3

conn = sqlite3.connect("company.db")
cursor = conn.cursor()

cursor.execute("DROP TABLE IF EXISTS Employee")

cursor.execute("""
CREATE TABLE Employee(
    Employee_ID INTEGER PRIMARY KEY,
    Employee_Name TEXT,
    Department TEXT
```

```
)  
""")  
  
cursor.executemany("""  
INSERT INTO Employee VALUES(?,?,?)  
""",[  
(1,"Rahul","CSE"),  
(2,"Anitha","ECE"),  
(3,"John","IT")  
)  
  
conn.commit()  
  
cursor.execute("SELECT * FROM Employee")  
  
rows = cursor.fetchall()  
  
for row in rows:  
    print(row)  
  
conn.close()
```

Output

```
(1, 'Rahul', 'CSE')  
(2, 'Anitha', 'ECE')  
(3, 'John', 'IT')
```

Question 4: SQLite Record Update

```
import sqlite3  
  
conn = sqlite3.connect("college.db")
```

```
cursor = conn.cursor()
```

```
cursor.execute("""
```

```
UPDATE Student
```

```
SET Department=?
```

```
WHERE Student_ID=?
```

```
""",("ECE",101))
```

```
conn.commit()
```

```
print("Record Updated Successfully")
```

```
conn.close()
```

Output

Record Updated Successfully

Question 5: SQLite Record Deletion

```
import sqlite3
```

```
conn = sqlite3.connect("bank.db")
```

```
cursor = conn.cursor()
```

```
cursor.execute("DROP TABLE IF EXISTS Customer")
```

```
cursor.execute("""
```

```
CREATE TABLE Customer(
```

```
    Customer_ID INTEGER PRIMARY KEY,
```

```
    Customer_Name TEXT
```

```
)
```

```
""")
```

```
cursor.executemany("""  
INSERT INTO Customer VALUES(?,?)  
""", [  
(101, "Rahul"),  
(105, "Arun"),  
(110, "David")  
])
```

```
conn.commit()
```

```
cursor.execute("""  
DELETE FROM Customer  
WHERE Customer_ID=?  
""", (105,))
```

```
conn.commit()
```

```
print("Record Deleted Successfully")
```

```
conn.close()
```

Output

Record Deleted Successfully

Question 6: SQLite Record Retrieval

```
import sqlite3
```

```
conn = sqlite3.connect("college.db")
```

```
cursor = conn.cursor()
```

```
cursor.execute("""
SELECT *
FROM Student
WHERE Student_ID=?
""",(101,))

row = cursor.fetchone()

if row:
    print(row)
else:
    print("Student Not Found")

conn.close()
```

Output

```
(101, 'Alice', 'ECE')
```

Question 7: SQLite Table Creation

```
import sqlite3

conn = sqlite3.connect("college.db")
cursor = conn.cursor()

cursor.execute("DROP TABLE IF EXISTS Student")

cursor.execute("""
CREATE TABLE Student(
    Student_ID INTEGER PRIMARY KEY,
    Name TEXT,
    Department TEXT
```

```
)  
""")
```

```
conn.commit()
```

```
print("Student Table Created Successfully")
```

```
conn.close()
```

Output

Student Table Created Successfully

Question 8: SQLite Record Insertion

Aim

To insert a new employee record into an SQLite database.

Theory

The INSERT statement is used to add new records to a table.

Code

```
import sqlite3
```

```
conn = sqlite3.connect("company.db")
```

```
cursor = conn.cursor()
```

```
cursor.execute("DROP TABLE IF EXISTS Employee")
```

```
cursor.execute("""
```



```
CREATE TABLE Employee(  
    Employee_ID INTEGER PRIMARY KEY,  
    Name TEXT,  
    Salary REAL  
)  
""")  
  
sql = ""  
INSERT INTO Employee(Employee_ID, Name, Salary)  
VALUES (?, ?, ?)  
""  
  
cursor.execute(sql, (101, "John", 55000))  
  
conn.commit()  
  
print("Record Inserted Successfully")  
  
cursor.close()  
conn.close()
```

Output

Record Inserted Successfully

Conclusion

The program successfully inserts a new employee record into the SQLite database.

Question 9: SQLite Record Update

Aim

To update the salary of an employee in an SQLite database.

Theory

The UPDATE statement modifies existing records in a table.

Code

```
import sqlite3

conn = sqlite3.connect("company.db")
cursor = conn.cursor()

cursor.execute("""
UPDATE Employee
SET Salary=?
WHERE Employee_ID=?
""", (65000, 101))

conn.commit()

print("Record Updated Successfully")

cursor.close()
conn.close()
```

Output

Record Updated Successfully

Conclusion

The program successfully updates the employee salary.

Question 10: SQLite Record Deletion

Aim

To delete all customer records belonging to Chennai.

Theory

The DELETE statement removes records from a table based on a condition.

Code

```
import sqlite3

conn = sqlite3.connect("bank.db")
cursor = conn.cursor()

cursor.execute("DROP TABLE IF EXISTS Customer")

cursor.execute("""
CREATE TABLE Customer(
    Customer_ID INTEGER PRIMARY KEY,
    Customer_Name TEXT,
    City TEXT
)
""")

cursor.executemany("""
INSERT INTO Customer VALUES(?,?,?)
""",[
(101,"Rahul","Chennai"),
(102,"Alice","Delhi"),
(103,"John","Chennai")
])
```

```
conn.commit()
```

```
city = "Chennai"
```

```
query = "DELETE FROM Customer WHERE City=?"
```

```
cursor.execute(query, (city,))
```

```
conn.commit()
```

```
print("Records Deleted Successfully")
```

```
cursor.close()
```

```
conn.close()
```

Output

Records Deleted Successfully